

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 39)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

- A A waste is assumed to cause pollution until research shows it is unlikely to do so ☐
- B The polluter must pay for any damage caused ☐
- C Emissions should be as low as reasonably achievable ☐
- D Only pollutants known to be toxic need to be controlled ☐

0 1 . 2

Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

.....

.....

.....

.....

0 1 . 3

Give **two** choices about emission location that reduce the severity of pollution.

[2 marks]

1

.....

2

.....

5

0 2

Table 1 compares a 1000 MW coal-fired power station with a 1000 MW nuclear power station.

Table 1

Feature	Coal-fired	Nuclear (uranium)
Fuel used per year	3.5 million tonnes	30 tonnes
CO ₂ emitted per kWh / g	900	12 (mostly fuel processing and construction)
Construction cost / £ billion	2	20
Fuel cost per kWh / pence	3.0	0.5
Typical response to changes in demand	Hours	Slow (days)

02 . 1

Explain how nuclear fission releases energy in a reactor.

[3 marks]

02 . 2

Use Table 1 to explain why nuclear power is used for base-load electricity but not for meeting peaks in demand.

[2 marks]

02 . 3

Use Table 1 to explain why nuclear power stations can be built where coal-fired stations could not.

[2 marks]

02 . 4

Explain why the CO₂ emissions of nuclear power are not zero.**[1 mark]**

02 . 5

Give **two** reasons why the true cost of nuclear electricity is uncertain.**[2 marks]**

1

2

10

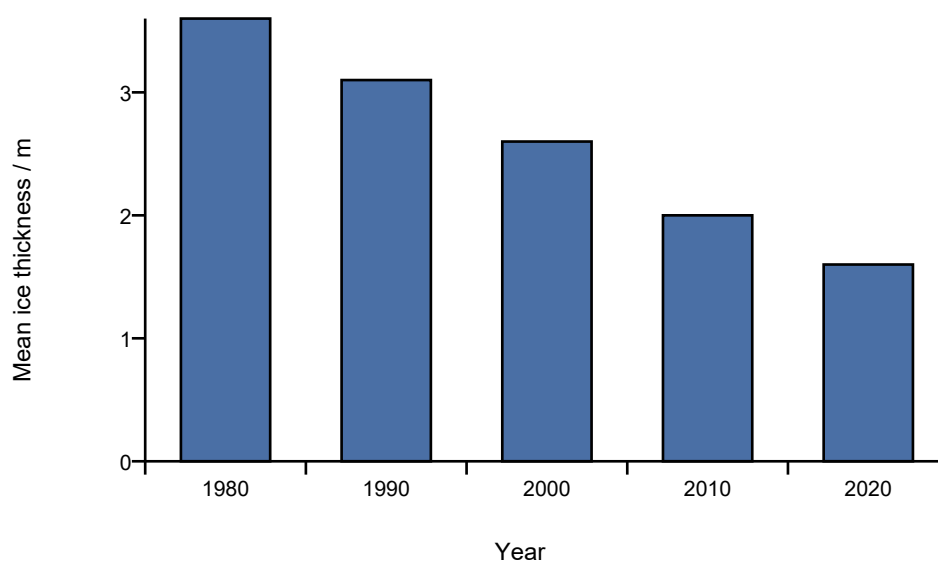
0 3

Table 1 shows the area of Arctic sea ice in September (the annual minimum) in selected years. Figure 1 shows the mean thickness of Arctic sea ice in the same years.

Table 1

Year	1980	1990	2000	2010	2020
September sea ice area / million km ²	7.7	6.2	6.3	4.9	3.9

Figure 1



0 3

1

Calculate the percentage decrease in September sea ice area between 1980 and 2020.

[2 marks]

Answer

%

03 . 2

Explain why the loss of sea ice does **not** directly raise sea level, but the loss of land ice does.

[2 marks]

03 . 3

Explain how the changes shown in Table 1 and Figure 1 could lead to further warming.

[3 marks]

03 . 4

Explain **two** ways in which rising temperatures cause sea level to rise.

[2 marks]

03 . 5

Explain why the break-up of an ice shelf can increase the rate at which land ice flows into the sea.

[1 mark]

10

0 4

Table 1 shows the pH of rainwater and the number of fish species in lakes in four areas of Scandinavia with different bedrock geology.

Table 1

Lake area	Bedrock	Mean rainfall pH	Mean lake pH	Number of fish species	Dissolved aluminium / mg dm^{-3}
A	Granite	4.3	4.6	0	0.42
B	Granite	4.8	5.1	3	0.20
C	Limestone	4.3	7.2	8	0.01
D	Limestone	5.5	7.6	9	0.01

0 4

1

State the pH below which rain is described as acid rain and explain why unpolluted rain is already slightly acidic.

[2 marks]

0 4

2

Use Table 1 to explain why lakes in area C contain many more fish species than lakes in area A, even though the rainfall has the same pH.

[4 marks]

0 4 . 3

Describe **two** effects of acid rain on non-living objects.**[2 marks]**

1

.....

2

.....

0 4 . 4

Explain how sulfur dioxide released from a power station is converted to sulfuric acid in the atmosphere.

[2 marks]

.....

.....

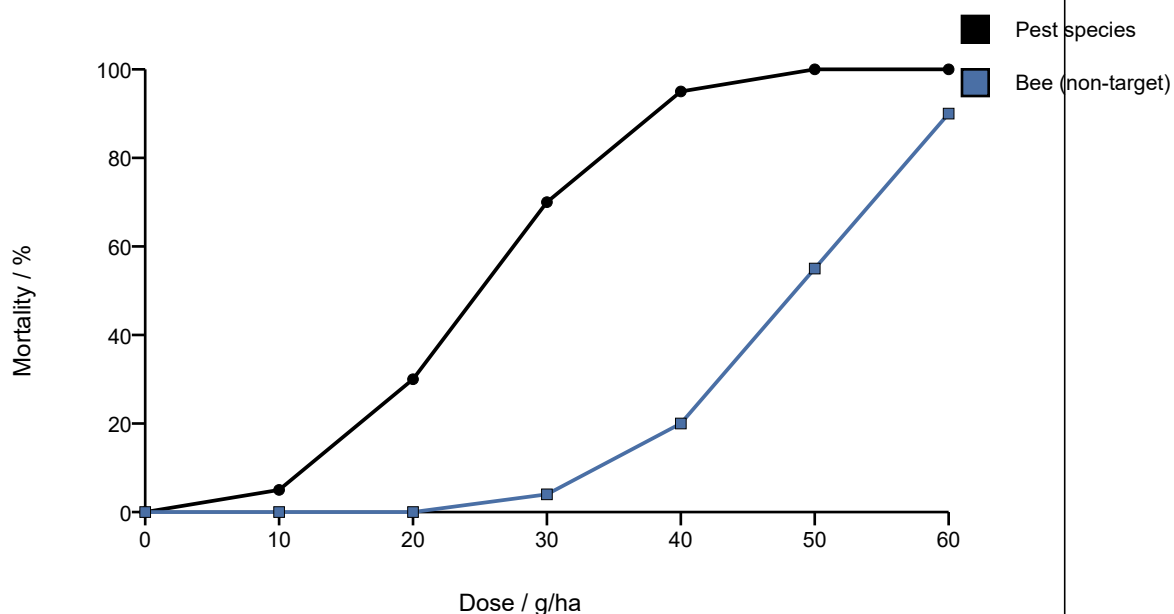
.....

.....

10

0 5

Figure 1 shows how the percentage mortality of a pest species and of a non-target bee species change with the dose of an insecticide applied to a crop.

Figure 1

0 5 . 1

Use Figure 1 to estimate the LD₅₀ of the insecticide for the pest species.**[1 mark]**

Answer g/ha

0 5 . 2 Define the term **MDNF** (maximum dose never fatal). **[1 mark]**

0 5 . 3 Use Figure 1 to explain why it may not be possible to kill all of the pest species without harming bees. **[3 marks]**

0 5 . 4 Suggest how the farmer could reduce the harm to bees without reducing the dose applied. **[2 marks]**

0 5 . 5 Explain why a pesticide with high specificity is less likely to cause environmental damage. **[3 marks]**

10

06 . 1

Explain why planting a new forest only sequesters carbon for a limited period, and how the storage can be extended.

[3 marks]

06 . 2

Compare pre-combustion and post-combustion methods of carbon capture.

[4 marks]

06 . 3

Describe **two** ways captured CO₂ can be stored and give **one** concern about long-term storage.

[3 marks]

10

07 . 1

Complete Table 1 by naming the most suitable statistical test for each hypothesis.

[4 marks]

Table 1

Hypothesis	Statistical test	
Noise levels decrease as distance from a road increases		
Fewer cars use a road after road charging is introduced (counts on 12 days before and after)		
Woodlands managed in different ways have different numbers of dormice		
Mean lobster mass is higher in areas where collection is banned		

07 . 2

Explain why the chi-squared test cannot be used to compare the dissolved oxygen concentrations of two rivers.

[2 marks]

07 . 3

Explain why a scientific investigation never 'proves' a hypothesis.

[2 marks]

07 . 4

Describe how error bars are added to a graph of means and how they are interpreted.

[2 marks]

10

08 . 1

Explain how integrated manufacture saves energy in the iron and steel industry.

[2 marks]

08 . 2

Explain why recycling is not always energy efficient.

[2 marks]

08 . 3

Explain how each of the following manages electricity infrastructure to conserve energy.

[4 marks]

High-voltage grid

.....

.....

.....

ICT management of supply and demand

.....

.....

.....

08 . 4

Give **two** locational factors affecting where new generating infrastructure is built.

[2 marks]

1

.....

2

.....

10

09 . 1

Explain **three** advantages of biofuels over other renewable energy resources.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

.....

09 . 2

Explain **three** disadvantages of biofuel crops.

[3 marks]

.....

09 . 3

Describe **two** new biofuel technologies.

[2 marks]

1

2

09 . 4

Explain how new low-temperature turbine technology could allow geothermal electricity generation in the UK.

[2 marks]

10

10 . 1

As high-grade mineral deposits become depleted, discuss the strategies that could be used to secure future supplies of metals.

[9 marks]

Do not write outside the box

Extra space

.....

.....

.....

.....

.....

.....

.....

1	0
---	---

 .

2

 State what is meant by the **stock** (resource base) of a mineral. **[1 mark]**

.....

.....

10

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
---	---	---	---

Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

OR

1 1 . 2

Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

END OF QUESTIONS